

Measurement of Localized Specific Absorption Rate (SAR) for Contactless Smartcard Readers Operating in the HF Band

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Abstract—This paper explores highly sensitive localized SAR measurement methods for contactless smartcard readers operating in the HF band, specifically 13.56 MHz. In this paper, the design and calibration of a low-frequency E -field probe is described. A method for synthesizing simulated human tissue using a technique developed for measurement of the conductivity of liquids at the HF band is also presented. Finally, both the near-field and localized specific absorption rates are measured for a contactless smartcard reader and compliance with Federal Communications Commission (FCC) maximum permissible exposure (MPE) and specific absorption rate (SAR) regulation is determined.

Index Terms—Contactless smartcard reader, electromagnetic exposure, specific absorption rate.

I. INTRODUCTION

A contactless smartcard is an intelligent card envisioned to replace currency and all types of cards including credit cards, telephone cards, driver's license, etc. Applications for contactless smartcards have increased rapidly over the past few years. A report on the smartcard industry by Dataquest Inc., a semiconductor industry analyst located in San Jose, CA, stated, "Although some standards issues, infrastructure issue, and software issues remain to be resolved, chip cards hold the promise of being one of the world's highest-volume markets for semiconductors." The report also indicated that 156 million smartcard microcontroller devices would be shipped in 1996, increasing to 990 million in the year 2000 [1]. A contactless smartcard is the size of a credit card. Unlike typical magnetic strip cards, contactless smartcards operate through RF energy transmission between the card and the reading device (reader). The smartcard reader provides the energy for the state machine or microprocessor in the smartcard through magnetic coupling. There is also an RF interface for data communication between the smartcard and reader as shown in [2]. Therefore, the electromagnetic exposure of the user from the reading apparatus must adhere to the recently introduced Federal Communications Commission (FCC) regulation limiting human exposure to RF emitted by FCC regulated transmitters.

Two basic parameters for the evaluation of RF energy exposure are maximum permissible exposure (MPE) and specific absorption rate (SAR). On August 1, 1996, the FCC adopted the 1992 ANSI/IEEE C95.1-1992, IEEE standard for safety levels with respect to human exposure to radio frequency electromagnetic fields: 3 kHz–300 GHz. The standard states maximum permissible exposure (MPE) limits of 1.2 A/m and 61.12 V/m. In cases where the device exceeds the MPE limits, it can be excluded from compliance with MPE limits if the localized SAR is below FCC limits. There are two types of SAR measurement defined: *whole body average* and *localized*. The whole body average is the SAR averaged over the entire body. Localized measurement is the SAR average over a small mass, typically a 1-g cube. There are also two limits stated: one for use in a *controlled* environment and another for the *uncontrolled*. The controlled environment is most generally defined as the exposure incurred knowingly by a person as part of employment. The uncontrolled environment is where the person has "no knowledge or control over exposure." The smartcard reader falls under the uncontrolled category with a localized SAR limit of 1.6 mW/gram [3].

SAR is defined as the time derivative of the incremental energy (dW) absorbed into an incremental mass (dm) [4]

$$\text{SAR} = \frac{d}{dt} \left(\frac{dW}{dm} \right).$$

In the context of smartcards, SAR is calculated as

$$\text{SAR} = \frac{\sigma |E|^2}{\rho} = c \frac{\Delta T}{\Delta t} \cdot [\text{mW/g}]$$

where

σ	conductivity of the tissue simulant [S/cm];
$ E ^2$	electric field strength [V^2/cm^2];
ρ	density of tissue simulant [g/cm^3];
c	specific heat capacity of tissue simulant [$\text{J}/\text{g}^\circ\text{C}$];
$\Delta T [^\circ\text{C}]$	change in temp when exposed for time Δt (s) [4].

Therefore, SAR can be determined from either electric-field amplitude or temperature measurement.

In the communication industry, SAR studies have focused on measurements at frequencies allocated for mobile communication usage, roughly 100 MHz–2 GHz. With the de-

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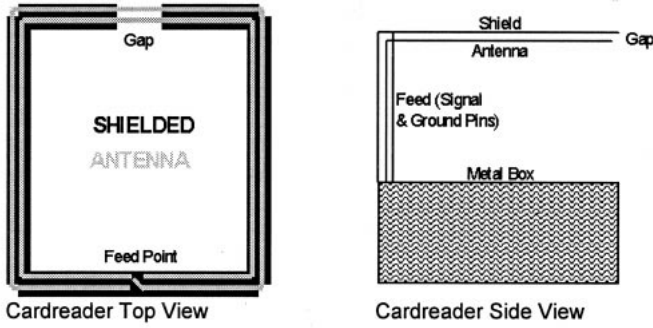


Fig. 1. Smartcard reader under test.

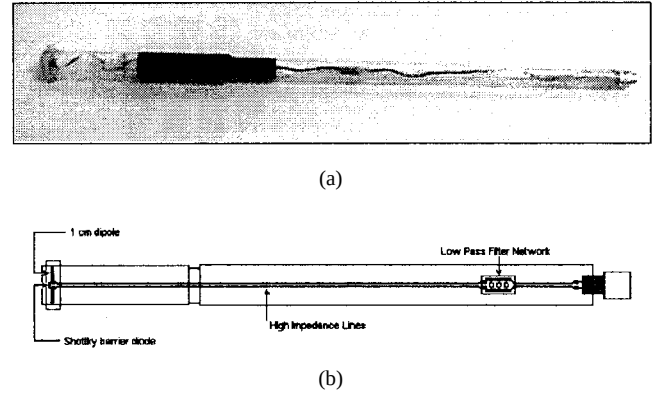
velopment of new applications, such as the smartcard operating outside of the common frequency range, different techniques must be introduced and improved for accurate SAR measurement. There have been only a few studies on SAR measurements in the frequency range of 1–30 MHz. One of these studies [5] focused on the measurement of whole-body SAR with the usage of full-sized human models placed in a large gradient layer calorimeter. Small nonperturbing probes are generally used for the determination of peak local SAR. The probe must have high accuracy and must be RF transparent so the measured fields are minimally perturbed. Temperature probes, like the thermistor-based BSD probe [6] meet these requirements and have been used previously, but they lack the sensitivity needed for measurements of most low-power products. Temperature probes also require a substantial exposure period to obtain a reliable temperature measurement. This condition drastically increases the scan time needed to find the position of peak SAR. *E* field probes give instantaneous readings and have high sensitivity, so they are preferred for SAR measurement. The *E*-field probes often consist of an electrically short dipole with a diode across its terminals and highly resistive lines to carry the detected signal for measurement [7]. Electrically short dipoles are preferred to tuned dipoles because their frequency response is very flat over a wide frequency range [8]. Often, three dipoles are configured in orthogonal orientation to provide an isotropic response for all polarizations of the *E* field.

II. EQUIPMENT UNDER TEST

The hardware under investigation for this RF exposure analysis is a prototype contactless smartcard reader (shown in Fig. 1). It consists of an approximately 9×9 cm two-turn square–spiral shielded loop located above a slightly larger ground plane. The antenna is implemented as a two-sided printed circuit board. A coaxial connector is mounted on the box for reliable connection to the signal generator/amplifier. For measurement purposes, a coil current necessary to perform a contactless smartcard transaction is used.

III. MEASUREMENT PROCEDURE

The method used to measure SAR at lower frequencies is basically identical to the one described in [9] and involves a robotic system to accurately position the *E*-field probe within

Fig. 2. *E*-field miniature dipole probe.

a human biological tissue simulant. Previous probes, designed for use at higher frequencies, lacked the sensitivity needed for lower frequency measurement, so a new single-axis probe with an increased dipole length was fabricated (Fig. 2).

The total dipole length was extended from 2.5 mm to 1 cm to increase the sensitivity roughly by a factor of 4, as indicated by the following [8]:

$$V_{OC}(t) = E_i(t)h_e$$

where

- $V_{OC}(t)$ the open-circuit voltage induced across the dipole gap;
- $E_i(t)$ the strength of the locally generated field;
- h_e the effective length of the dipole.

Using a robotic positioning system, the probe scans the area of interest in the simulated tissue and the SAR is determined using the probe output and a conversion factor (CF). The CF is determined during the calibration procedure described in detail later in this report.

IV. DIELECTRIC PROPERTIES MEASUREMENT OF TISSUE SIMULATING MIXTURE

One difficulty was the determination of the chemical composition of the proper human tissue simulant, whereby the SAR measurements can take place. A simulation of soft muscle tissue representative of the human torso was selected for this measurement. At higher frequencies (150–1500 MHz), the dielectric properties are determined by measuring the change in amplitude and phase of the S_{21} parameter at various positions along a 30-cm slotted short-circuited coaxial line using a network analyzer, [10]. The long wavelength at 13.56 MHz poses a problem of reflections from the end of the slotted coaxial line when measurements of amplitude and phase are taken. One solution to this problem is a longer line but such a structure would be impractical. Therefore, a different approach was taken.

The conductivity of the tissue simulant (liquid) is determined using the change in the real part of the admittance of an open-ended coax line as it is incrementally filled. In effect, the change in admittance of the coax is measured as the added test mixture incrementally replaces the air dielectric, as shown in Fig. 3. The open-ended coax has a shield radius of 7.27

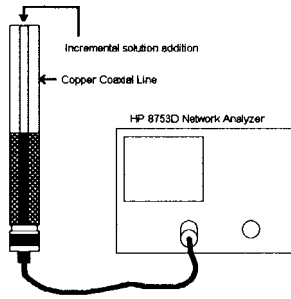


Fig. 3. Solution conductivity measurement setup.

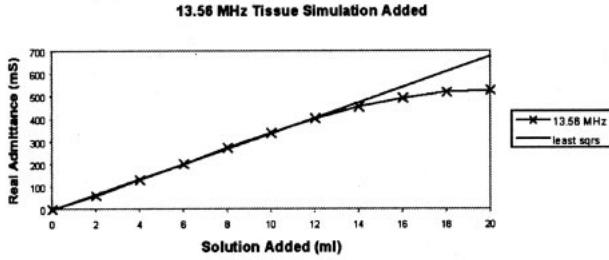


Fig. 4. Plot of admittance of incrementally filled coaxial line.

mm and an inner conductor radius of 3.15 mm. The coax is 20 cm in length. Fig. 4 shows the complex impedance and the least squares straight line approximation for the change in the real part impedance as 0.5% NaCl solution is added to the coaxial line.

The slope of the least-squares straight-line approximation was used in the following equation to determine the conductivity of the solution.

For the incrementally filled coaxial line

$$C = \text{Incremental capacitance of the coaxial line} \\ = \frac{2\pi\epsilon_{rc}}{\ln(b/a)}; \left[\frac{F}{m} \right]$$

$$Y_{\text{Coax}} = \text{Admittance of the coaxial line} = j\omega C; [11].$$

If a coaxial line of length L_{TOT} is filled to length L_{SOL} with a dielectric of complex permittivity ϵ_{rc} , then

$$Y_{\text{TOT}} = \text{Admittance of the coaxial line} \\ = j\omega \frac{2\pi\epsilon_0}{\ln(b/a)} [L_{\text{SOL}}(\epsilon_{rc} - 1) + L_{\text{TOT}}]$$

where

- b/a ratio between the outer and inner conductor of the coax;
- L_{TOT} total length of the coaxial line;
- L_{SOL} part of the line filled with solution;
- ϵ_{rc} complex relative permittivity $= (\epsilon_r - j\frac{\omega\epsilon'' + \sigma'}{\omega\epsilon_0})$;
- σ_e $\sigma' + \omega\epsilon'' = \text{real effective conductivity}; [12].$

Substitute for ϵ_{rc}

$$Y_{\text{TOT}} = j\omega \left(\frac{2\pi\epsilon_0}{\ln(b/a)} \right) \left[\left(\epsilon_r - 1 - \frac{j\sigma_e}{\omega\epsilon_0} \right) L_{\text{SOL}} + L_{\text{TOT}} \right].$$

The real part of the admittance after simplification is

$$Y_{\text{RE}} = \frac{2\pi\sigma_e}{\ln(b/a)} L_{\text{SOL}}.$$

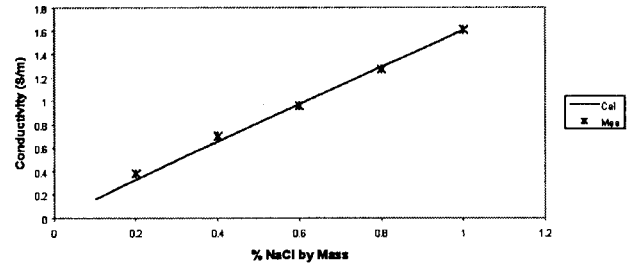


Fig. 5. Calculated and measured conductivity of saline solutions at 13.56 MHz.

Taking the derivative and solving for σ_e

$$\sigma_e = \frac{dY_{\text{RE}}}{dL_{\text{SOL}}} \left(\frac{\ln(b/a)}{2\pi} \right)$$

where

$\frac{dY_{\text{RE}}}{dL_{\text{SOL}}}$ = the slope of the least squares approximation of the change in Y_{RE} as solution is added.

Various saline mixtures were tested using this technique and the results are shown in Fig. 5. The theoretical calculation of the dielectric properties of saline, based on revisions of [13] made by Dr. C. Davis in July 1987, are also plotted to show good correlation.

At 13.56 MHz, soft muscle tissue has a dielectric constant of 132.1 and a conductivity of 0.66 S/m [14]. The dielectric constant of the simulated tissue solution could not be determined using this technique. Though the dielectric constant can be simply derived using the imaginary change in admittance, the experimental results were totally reliable. At this frequency, only penetrating magnetic fields (there are no external incident electric fields of relevance) induce electric fields in the mixture and, therefore, conductivity is the important parameter in mixture synthesis. In addition, the lower relative dielectric constant of the simulated tissue ($\epsilon_r \approx 70$ versus 132.1) enhances the penetration of the normal component of external E fields. In the absence of incident tangential E fields, as in this case, the experimental mixture provides a slight overestimate of the SAR in tissue from external near field RF sources. Fig. 5 shows that the 0.38% NaCl solution has a conductivity that closely matches the soft muscle tissue conductivity of 0.66 S/m at 13.56 MHz. In addition to proper conductivity, low specific heat is also desired. This is to lower the amount of energy needed to obtain measurable temperature rise. The addition of sugar into the solution lowers the specific heat. The addition of HEC (hydroxyethyl cellulose) increases the viscosity of the simulated tissue. High viscosity is needed to reduce effects of convective cooling. The final mixture concentrations were as follows:

13.56 MHz Mixture

Concentrations by mass

1	Deionized H ₂ O	63.60%
2	Sugar	35%
3	NaCl	0.80%
4	HEC (Jelling agent)	0.20%
5	Bactericide	0.40%

(NaCl concentrations had to be increased from the saline to offset the effects of the HEC and the bactericide.)

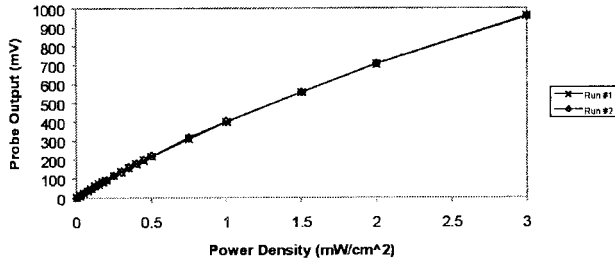


Fig. 6. Low-frequency E -field probe response in TEM cell; 13.56 MHz.

V. PROBE CALIBRATION

The first step of calibration is performed in the known uniform field of a Crawford TEM cell [15]. The probe output is measured for varying power densities inside the cell (Fig. 6). This is to eliminate any future variation in probe sensitivity. Though the sensitivity of the dipole itself is constant, the characteristics of the preamplifier and high-impedance lines appear to slightly change the overall sensitivity of the system over time. Before testing, the probe is first placed in the TEM cell and the response is normalized to the initial TEM cell calibration.

The second part of the calibration is known as the thermal calibration. As stated above, SAR can be calculated using the equation $SAR = c \frac{\Delta T}{\Delta t}$ involving the measurement of temperature rise in a tissue simulant. The other equation stated above $SAR = \frac{\sigma |E|^2}{\rho}$ cannot be used without the difficult determination of $|E|^2$ inside of the simulated tissue solution [16]. The calibration is performed at high RF power levels where a measurable temperature increase is obtainable. The temperature response, at the high power levels, are linearly extrapolated to the lower power levels where the probe output is proportional to $|E|^2$. The slopes of both the probe output and temperature with respect to change in power are found. The ratio of these slopes multiplied by the specific heat and divided by the exposure time to determine the CF. This factor allows for the conversion from probe output to SAR.

The calibration is performed in a flat phantom containing the proper simulated human tissue. At higher frequencies, the phantom is illuminated by a tuned half-wave dipole and the 30-s temperature rise is correlated to the probe output at the position of highest field strength for varying dipole output powers by a CF. At 13.56 MHz, the calibration is still performed in a flat phantom using a tuned loop designed to produce the required fields. The antenna used for testing consists of a 30-cm-diameter loop that is tuned with three capacitors and two inductors to a frequency of 13.56 MHz. The inductors and loop are made from 0.23-cm-diameter wire, and the capacitor plates are from 7.5-cm-wide copper plates, as shown in Fig. 7. The air gap capacitor plates are formed with Teflon spacers and clamps to avoid arcing at high power levels. This arrangement allowed the antenna to be nearly lossless with a quality factor greater than 500 when away from lossy media and a quality factor of 330 when dissipative materials such as a simulated human body are in close proximity to the antenna. Return loss bandwidth is 0.043 MHz at the -10 dB level in free-space. When next to the human phantom,

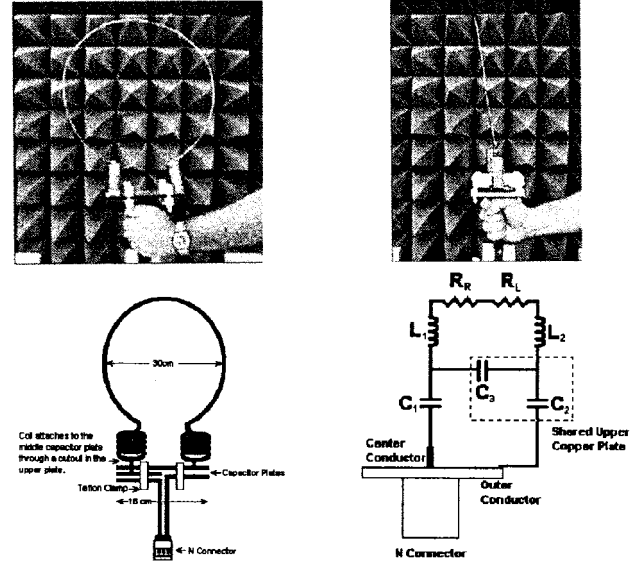


Fig. 7. Loop antenna and equivalent circuit.

the bandwidths are 0.032 MHz and 0.101 MHz at the -6 -dB points and -10 -dB points, respectively.

In the equivalent circuit of the loop

- R_R radiation resistance of the loop;
- R_L loss resistance of the loop and coils;
- L_1 L_2 = inductance of the coils and loop;
- C_1, C_2 capacitance of series capacitors (C_1 and C_2 should be as close as possible in value for balancing);
- C_3 shunt capacitance.

(C_2 and C_3 share the upper plate.)

For calibration, the loop is placed directly under the flat phantom of tissue simulant with the plane of the loop parallel to the bottom of the phantom. Since the probe measures along only one axis, two perpendicular probe scans in the horizontal plane are taken in the mixture and summed to obtain a complete field scan. The point of maximum field strength is determined.

First, the BSD temperature probe is placed at the point of maximum field strength and temperature measurements are taken at various powers for a 1-min duration. The BSD temperature probe is provided and calibrated by the BSD Medical Corporation, and measurements were taken with a four-input Ωm through a GPIB interface. The BSD thermometer has a thermistor sensor and carbonized Teflon feed lines, which are RF transparent. The thermometer, in a stable temperature environment, has a reliable sensitivity of 0.01°C .

Measurements were taken at 1-s intervals for a minute. The linear increase in the temperature in Fig. 8 indicates minimal effect of heat diffusion in the mixture. A least-squares linear approximation is performed to find the change in temperature per 30 s of exposure. Then the temperature increase is plotted as a function of power, as shown in Fig. 9.

A least-squares straight-line approximation is made of the data points and the slope of this line is denoted as $\frac{dT_{mea}}{dP_{mea}}$ and is found to be 0.001174°C/W .

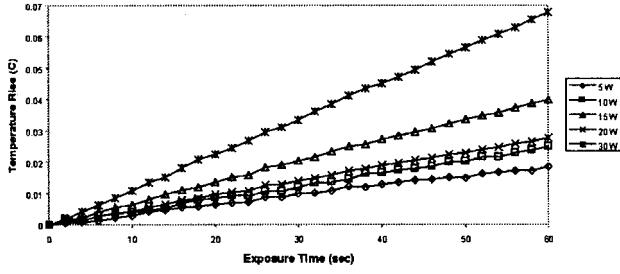


Fig. 8. Sixty second temperature linearity at different powers.

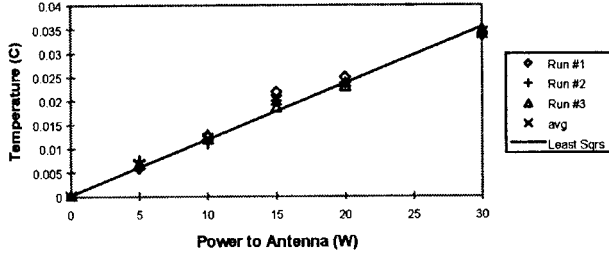
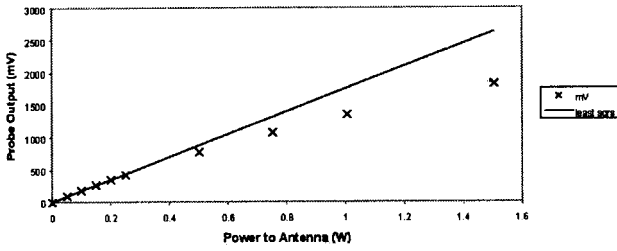


Fig. 9. Thirty second temperature response.

Fig. 10. *E*-field probe response.

The *E*-field probe is then placed at the position of max exposure; the probe reading is maximized to assure sensor alignment with the *E* field and measurements are taken at increasing power levels. Fig. 10 shows the *E*-field probe output as the power is increased.

The least squares straight-line approximation is found for what is known as the square-law region of the probe response. The square-law region is the region where the probe output is directly proportional to $|E|^2$ and is evident as the region where the probe output closely follows the straight-line approximation. The slope of the straight-line is denoted as, $\frac{dE_{mea}}{dP_{mea}}$ and is found to be 1751.33 mV/W and the *y* intercept can be assumed to be zero. Then the two slope values are used to determine the CF 1

$$CF = \text{Conversion Factor} = \frac{\frac{dT_{mea}}{dP_{mea}} \cdot C}{\frac{dE_{mea}}{dP_{mea}} \cdot \Delta t}; \left[\frac{W}{g \cdot mV} \right]$$

where

- $\frac{dT_{mea}}{dP_{mea}}$ change in temperature in Δt exposure time per unit power increase. $[\frac{^\circ C}{W}]$;
- C specific heat of the simulated tissue $[\frac{J}{g \cdot ^\circ C}]$;
- $\frac{dE_{mea}}{dP_{mea}}$ change in probe output voltage per unit power increase $[\frac{mV}{W}]$;
- ΔT exposure time [s].

The tissue simulant has a specific heat of $3.0 \frac{J}{g \cdot ^\circ C}$.

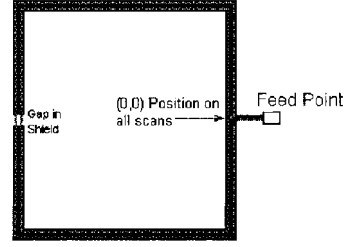
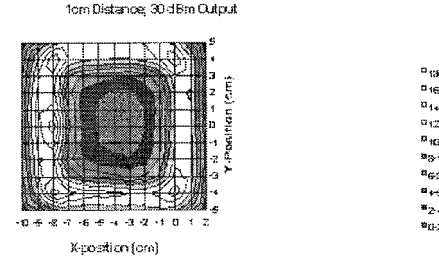


Fig. 11. Orientation of the smartcard reader in the following scans.

Fig. 12. Smartcard reader free-space *H*-field scan (A/m).

In this experiment, CF was determined to be $0.000067034 \frac{mW}{g \cdot mV}$.

Then one can see that the SAR can be calculated by the following:

$$SAR = \text{Probe output} \times CF; \left[\frac{mW}{g} \right].$$

For measurement of fields where the probe response extends beyond the square-law region, the diode saturation must be considered. This response causes an underestimation of the measurement. To eliminate this problem, the relationship between probe response and power is approximated with a third-order polynomial

$$P_{comp} = 1.86 \times 10^{-11} E_{mea}^3 + 2.57 \times 10^{-7} E_{mea}^2 + 5.11 \times 10^{-4} E_{mea} + 4.55 \times 10^{-4}.$$

The " P_{comp} " value is then used in the point slope equation

$$E_{comp} = \frac{dE_{mea}}{dP_{mea}} P_{comp}.$$

E_{comp} is the diode compensated probe response and is used for SAR measurement.

VI. MEASUREMENT RESULTS

Both a free-space and two SAR measurements were performed on the smartcard reader. In all three scans, the feed point of the loop is located at the origin of the coordinate system, point (0,0) as shown in Fig. 11. The free-space measurement was performed using the system described in [17] and a two-axis miniature loop *H*-field probe.

The measurement is shown in Fig. 12. It is evident here that the location of max *H* field in the near field closely follows the coils of the reader. The measured value of 18.87 A/m exceeds the FCC MPE limit of 1.2 A/m.

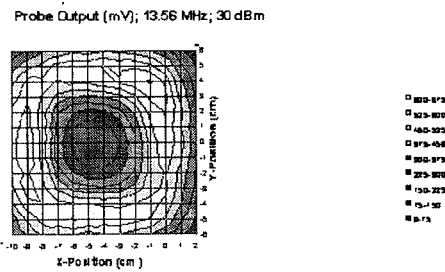


Fig. 13. Smartcard reader area scan in the flat phantom.

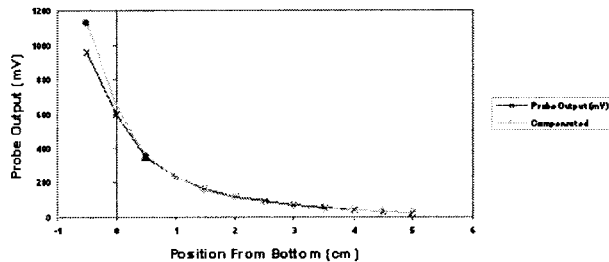


Fig. 14. Probe output (mV) for SAR measurement.

The smartcard reader was placed under a rectangular flat-bodied phantom of simulated tissue and measurements along both the x and y axis were taken. These measurements were summed and shown in Fig. 13. From these measurements, it is evident that the region of high E -field penetration into the solution closely follows the distribution of H fields from the loop of the reader coil. The position of maximum field strength position is at the corner of the coil.

At the point of maximum field strength, the probe measures the decay of the E field as it retracts from the bottom, as shown in Fig. 14.

The box marks on the figure show actual measured data taken in 5-mm increments. The data is compensated for the diode response and shown as the "o" marks. As expected, these values are slightly higher than the measured data when the probe is measuring the higher fields close to the surface. The dipole is offset from the bottom of the probe so the E field directly on the bottom of the phantom is predicted by extrapolation and is shown by the filled circular mark on the graph. The filled triangular mark shows the E field at 1-cm distance from the bottom of the phantom. The probe outputs at these two points are averaged to determine the highest 1-g average probe output. Peak spatial SAR is determined by multiplying this value by the CF. The peak spatial gram averaged SAR of the contactless smartcard reader is determined to be 0.055 mW/g.

VII. CONCLUSION

This paper shows the different techniques that had to be explored to adequately measure a device at 13.56 MHz. These techniques ranged from the manufacture and calibration of a probe designed for lower frequencies and a new method for conductivity measurements. Though the smartcard reader exceeded the FCC limits for MPE, the SAR measurement of

0.055 mW/g shows it is much lower than the FCC SAR limit of 1.6 mW/g. The SAR measurement would take precedence over MPE limits to determine compliance.

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